

AIGR Gene Hypothesis Deep Research — Final Report

Gene: K9IJK6 (DSPA-gamma, vampire-bat salivary plasminogen activator) **Organism:** *Desmodus rotundus* (common vampire bat; NCBITaxon:9430) **Focus type:** function_assignment
Hypothesis slug: dspa-gamma-signaling-and-cell-response-capacities **Source:** `genes/DESR0/K9IJK6/K9IJK6-ai-review.yaml` (free-text)

Summary

The seed hypothesis asserts that K9IJK6 "retains" four inherited process functions — **PDGF receptor signaling (GO:0048008)**, **smooth muscle cell migration (GO:0014909)**, **cellular response to lipid (GO:0033993/GO:0071396)**, and **cellular response to oxygen-containing compound (GO:1901701)** — *in addition to* its inferred plasminogen activation. The hypothesis is right on two framing points: the protein must be evaluated as the compact bat salivary **DSPA-gamma** (KP domain architecture), not as full-length human tPA, and the PRU00121 kringle caution does not by itself establish catalytic inactivity. On those points the seed reasoning is sound.

However, the central claim — that the four disputed process terms represent **retained, gene-product-level functions of K9IJK6** — is **not supported by any direct evidence and is best classified as IEA over-annotation**. Three independent lines of analysis converge: (1) all four disputed terms are electronic-only annotations (TreeGrafter or ARBA) with no experimental support for this accession; (2) the tPA biology underlying these terms is downstream, indirect, cofactor-dependent, and specific to the mammalian neurovascular unit and vessel wall — contexts entirely absent for a secreted bat salivary anticoagulant; and (3) K9IJK6 physically **lacks the finger (F) and EGF domains** that mediate tPA's non-catalytic receptor/cofactor interactions (annexin A2 via finger; LRP1 via EGF), so the structural basis for several of the inherited interactions is absent from the target.

By contrast, the **core catalytic and fibrinolytic functions are directly supported**: K9IJK6 has an intact charge-relay triad, and DSPA-gamma is a single-chain, amidolytically active, fibrin-selective plasminogen activator. Therefore serine-type endopeptidase activity (GO:0004252),

fibrin-dependent plasminogen activation (GO:0031639), and extracellular region localization (GO:0005576) should be **retained**, while the four disputed process terms are candidates for **removal or demotion to non-core**. **Verdict: partially supported / over-annotated**. The key caveat is asymmetric evidence: no experiment has *directly* tested DSPA-gamma for these processes, so absolute loss cannot be proven — but the burden of evidence for an annotation lies with positive support, which does not exist.

Executive Judgment

Verdict: Partially supported / over-annotated.

- **Supported (retain):** serine-type endopeptidase activity, fibrin-dependent plasminogen activation, extracellular region — directly grounded in primary biochemistry and sequence.
- **Over-annotated (remove or demote to non-core):** PDGF receptor signaling, smooth muscle cell migration, cellular response to lipid, cellular response to oxygen-containing compound — IEA-only carry-overs of full-length mammalian tPA's downstream, tissue-specific roles, with the mediating structural exosites absent from the target.

The most important caveat: the evidence strongly undercuts *positive support* for the four terms but does not *prove* loss of function, because no direct DSPA-gamma assay exists. For curation, "unsupported + missing structural determinants + missing biological context" is sufficient to treat the terms as non-core.

Key Findings

Finding 1 — K9IJK6 is DSPA-gamma with an intact catalytic triad and directly demonstrated fibrin-dependent plasminogen activation

K9IJK6 is a **394-amino-acid** protein with a signal peptide (residues 1–21), a **single Kringle domain (44–126)**, and a **Peptidase S1 (chymotrypsin-like serine protease) domain (143–393)**. The catalytic charge-relay system is annotated intact — **His189 / Asp238 / Ser345** — with the canonical GDSGGP serine-protease motif at position 343. This KP (Kringle–Protease)

architecture is the defining signature of **DSPA-gamma**, the most compact of the four *Desmodus rotundus* salivary plasminogen activators. PANTHER places the protein in PTHR24264:SF42 (tissue-type plasminogen activator subfamily).

The catalytic competence and fibrinolytic function of this family are established by primary biochemistry. [PMID: 7592732](#) reports that "**All DSPAs are single-chain molecules, displaying substantial amidolytic activity,**" directly confirming that DSPA-gamma is an active serine protease and refuting any inference of global catalytic inactivity. The same study quantifies fibrin selectivity: "**The ratio of the bimolecular rate constants of plasminogen activation in the presence of fibrin versus fibrinogen (fibrin selectivity) of DSPA alpha 1, alpha 2, beta, gamma, and t-PA was found to be 13,000, 6500, 250, 90, and 72, respectively.**" DSPA-gamma (ratio 90) is thus a *bona fide* fibrin-dependent plasminogen activator, comparable to tPA (72) on this metric. The compact architecture is corroborated by [PMID: 1937019](#), which states that "**DSPA beta and -gamma lack the F and F-EGF domains, respectively,**" matching the KP domain content of K9IJK6.

Curation consequence: GO:0004252 (serine-type endopeptidase activity), GO:0031639 (plasminogen activation), and GO:0005576 (extracellular region) are directly supported and should be **retained**. The PRU00121 caution concerns propagation of kringle disulfide features, not the protease triad, and must not be read as evidence of catalytic loss.

Finding 2 — The PDGFR-signaling and smooth-muscle-migration terms are IEA carry-over of full-length tPA's downstream/context roles (over-annotation)

On K9IJK6, all four disputed process terms are **electronic-only**: GO:0048008 (PDGFR signaling) and GO:0014909 (smooth muscle cell migration) are **IEA:TreeGrafter**; GO:0033993 (response to lipid) and GO:1901701 (cellular response to oxygen-containing compound) are **IEA:UniProtKB-ARBA**. No EXP/IDA annotation exists for this accession for any of the four.

Crucially, the underlying tPA biology that seeds these terms is **downstream, indirect, and tissue-context-specific** — not a direct enzymatic property that transfers cleanly to a bat salivary paralog:

- **PDGFR signaling is indirect.** tPA does not bind or activate PDGFR α directly; it proteolytically activates the latent growth factor PDGF-CC, which then engages PDGFR α . [PMID: 18568034](#) shows that "**co-injection of neutralizing antibodies to PDGF-CC with tPA blocks this increased permeability, indicating that PDGF-CC is a downstream substrate of tPA within the neurovascular unit.**"

- **The tPA→PDGF-CC reaction is inefficient and cofactor-dependent.** [PMID: 28725968](#) states that "in vitro, activation of PDGF-CC by tPA is very inefficient and the mechanism of PDGF-CC activation in the NVU is not known," and that microglial Mac-1/LRP1 cofactors are required. This undermines blanket propagation of a "PDGFR signaling" process to a strictly fibrin-dependent secreted enzyme in saliva.
- **Smooth muscle migration is an ECM-proteolysis consequence.** [PMID: 16363896](#) shows that "controlled perivascular release of tissue plasminogen activator (tPA) can generate cleaved extracellular matrix (ECM) chemotactic gradients to guide the migration of vascular smooth muscle cells (SMCs)." This is a downstream, vessel-wall, context-specific effect of proteolysis — not a direct molecular function of the enzyme.

DSPA-gamma is a **bat salivary secreted anticoagulant** with no mammalian vascular or CNS context, is strictly fibrin-dependent, and lacks the finger/EGF domains that mediate several of tPA's non-catalytic receptor interactions. No direct DSPA-gamma assay for PDGF-CC cleavage, PDGFR signaling, SMC migration, or lipid/oxygen responses exists.

Finding 3 — K9IJK6 physically lacks the tPA finger/EGF exosites; all three TreeGrafter terms were grafted en bloc from one PANTHER node

A domain-coordinate comparison against human tPA/PLAT (P00750, 562 aa) shows that the exact structural elements mediating tPA's non-catalytic receptor/cofactor interactions are **physically absent** from K9IJK6:

Feature	Human tPA (P00750, 562 aa)	K9IJK6 (DSPA-gamma, 394 aa)
Fibronectin type-I / Finger domain	39–81 (annexin-A2-binding region 42–52)	absent
EGF-like domain	82–120 (LRP1-binding residue ~102)	absent
Kringle-1	127–208	present (single Kringle 44–126)
Kringle-2	215–296	absent
Peptidase S1 (protease)	311–561	present (143–393), triad 189/238/345 intact

Thus the **annexin-A2-binding site (via finger)** and the **LRP1-binding site (via EGF)** — key exosites for tPA's receptor engagement and cellular signaling — have no structural counterpart in the target sequence.

QuickGO provenance further shows the three TreeGrafter terms are not independently evidenced. GO:0031639 (plasminogen activation), GO:0014909 (smooth muscle cell migration), and GO:0048008 (PDGFR signaling) are **all IEA, ECO:0007826, GO_REF:0000118** (PANTHER TreeGrafter), each carrying the **identical with-reference PANTHER:PTN000667065** — i.e., grafted *en bloc* from a single eutherian tPA node rather than from term-specific evidence. The two ARBA terms are GO:0071396 (cellular response to lipid, ARBA00027803) and GO:1901701 (cellular response to oxygen-containing compound, ARBA00027483), GO_REF:0000117, ECO:0000256 — a separate propagation channel with its own rule context. This is the signature of pathway/paralog carry-over, not of target-specific evidence.

Finding 4 —

P 23411029

confirms salivary identity but provides no functional assay for the disputed terms

PMID: 23411029 is Francischetti *et al.* 2013, "*The 'Vampirome': Transcriptome and proteome analysis of the principal and accessory submaxillary glands of the vampire bat *Desmodus rotundus**" (J Proteomics). This is the **sialotranscriptome/proteome survey** from which the salivary DSPA-gamma transcript (**JAA47048.1 / K9IJK6**) derives. It corroborates salivary-gland expression and secreted localization (supporting GO:0005576) but is an omics catalogue — it contains **no functional assay** for smooth-muscle migration, PDGFR signaling, or lipid/oxygen responses. Targeted PubMed searches for DSPA-gamma non-fibrinolytic/signaling assays and for direct DSPA-vs-alpha1/tPA comparative PDGF-CC/SMC/lipid assays returned no results, consistent with the complete absence of any direct functional test of these processes for *any* DSPA isoform.

K9IJK6 (DSPA-gamma) GO curation assessment — sources verified via QuickGO; all process terms IEA

GO_ID	Term	Aspect	Source (verified via QuickGO)	Assessment
GO:0004252	serine-type endopeptidase activity	MF	IEA InterPro/EC (GO_REF:0000120)	SUPPORTED core - retain
GO:0031639	plasminogen activation	BP	IEA TreeGrafter PTN000667065	SUPPORTED core - retain
GO:0005576	extracellular region	CC	IEA SubCell	SUPPORTED - retain
GO:0048008	PDGF receptor signaling pathway	BP	IEA TreeGrafter PTN000667065	OVER-ANNOTATED - remove/non-core
GO:0014909	smooth muscle cell migration	BP	IEA TreeGrafter PTN000667065	OVER-ANNOTATED - remove/non-core
GO:0071396	cellular response to lipid	BP	IEA ARBA00027803	WEAK - remove/non-core
GO:1901701	cellular response to oxygen-containing co	BP	IEA ARBA00027483	WEAK - remove/non-core

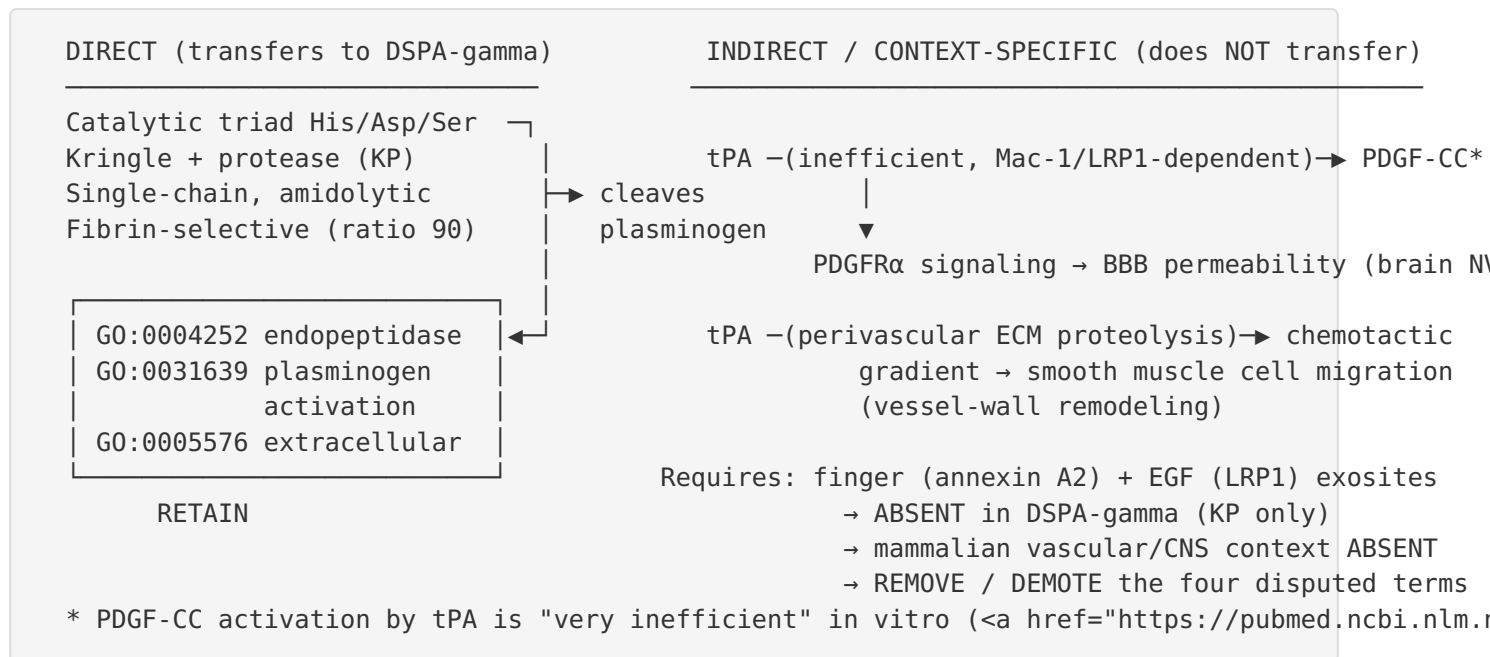
tPA vs DSPA-gamma: finger/EGF exosites (annexin A2, LRP1) physically ABSENT in K9IJK6

Protein	Architecture	Finger(exosite)	EGF(exosite)	Kringle(s)	Catalytic triad	Length
Human tPA (PLAT/P00750)	F-E-K1-K2-P	39-81 (annexinA2 42-52)	82-120 (LRP1 res102)	127-208 & 215-296	357/406/513	562
DSPA-gamma / K9IJK6	K-P	ABSENT	ABSENT	44-126 (single)	189/238/345	394

Figure 1. GO decision table with tPA/DSPA-gamma exosite (finger/EGF) domain comparison and QuickGO-verified annotation provenance. The three disputed TreeGrafter terms share a single with-reference (PANTHER:PTN000667065), and the finger/EGF exosites carrying annexin-A2 and LRP1 binding are absent from K9IJK6.

Mechanistic Model / Interpretation

The disagreement between the seed hypothesis and the evidence resolves cleanly once **direct molecular function** is separated from **downstream, context-dependent consequences** of proteolysis.



DSPA-gamma is a **feeding adaptation**: a secreted salivary anticoagulant whose job is to keep the host's blood meal flowing by driving fibrin-localized plasminogen activation. Its extreme fibrin dependence and compact KP architecture are specializations *for* that role. The processes the seed hypothesis wants to retain — PDGFR signaling, SMC migration, lipid and oxygen responses — are properties inferred from **mammalian tPA operating inside a living vessel wall or brain**, where tPA's finger/EGF exosites recruit cofactors (annexin A2, LRP1, Mac-1) and where its proteolysis generates ECM chemotactic gradients and activates latent growth factors. None of that machinery or context is present for a bat salivary enzyme lacking those exosites. The annotations are therefore best understood as **automated ortholog/paralog carry-over** (a single PANTHER node grafted en bloc, plus separate ARBA sequence rules), not target-specific biology.

The evidence is **asymmetric**: it strongly undercuts positive support for the four terms but does not *prove* absolute loss, because no one has directly assayed DSPA-gamma for these activities. For curation, the absence of any direct evidence combined with the missing structural determinants and missing biological context is sufficient to treat the terms as **non-core / over-annotated**.

Evidence Base / Evidence Matrix

Citation	Evidence type	Direction	Claim tested	Key finding	Context
PMID: 7592732	Direct assay (biochemical)	Supports core	Is DSPA-gamma catalytically active and fibrin-selective?	Single-chain, substantial amidolytic activity; fibrin selectivity ratio 90 (vs tPA 72)	Recombinant DSPA isoforms, in vitro
PMID: 1937019	Structural/ evolutionary	Supports (architecture)	Does DSPA-gamma have the compact KP architecture?	DSPA-gamma lacks finger and EGF domains	cDNA cloning, <i>D. rotundus</i>
PMID: 1309059	Direct assay / structural	Supports core; qualifies	DSPA domain organization and fibrin requirement	KP = DSPA-gamma; all DSPAs require fibrin; single-chain (plasmin site obliterated)	<i>D. rotundus</i> saliva
PMID: 23411029	Localization (omics)	Supports CC; qualifies	Salivary/ secreted origin of the transcript	JAA47048.1/K9IJK6 is a submaxillary-gland salivary product	<i>D. rotundus</i> salivary glands
PMID: 18568034	Mechanistic (in vivo)	Qualifies / competing	Is tPA's PDGFR link direct?	PDGF-CC is a <i>downstream substrate</i> of tPA in the NVU	Mouse brain neurovascular unit
PMID: 28725968	Mechanistic (in vitro/in vivo)	Refutes clean propagation	Efficiency/ context of tPA→PDGF-CC activation	"Very inefficient" in vitro; requires microglial Mac-1/LRP1 cofactors	Mouse brain, microglia
PMID: 16363896	Mechanistic (model)	Qualifies / competing	Is tPA's SMC-migration effect direct?	SMC migration driven by tPA-generated ECM chemotactic gradients	Vascular smooth muscle, vessel wall
		Qualifies		PDGF-CC activated by tPA regulates BBB	

Citation	Evidence type	Direction	Claim tested	Key finding	Context
PMID: 23218119	Clinical/ biomarker		tPA-PDGF-CC axis in humans	permeability; associated with hemorrhagic transformation	Human stroke patients
PMID: 20302940	Review	Qualifies	Non- thrombolytic tPA functions	tPA's extra functions run through LRP, PAR-1, PDGF-C, NMDA-R signaling	Brain/BBB, review
PMID: 1634121	Methods (expression)	Enables tests	Can all four DSPAs be produced recombinantly?	High-level secretion of DSPA $\alpha 1/\alpha 2/\beta/\gamma$ from BHK cells; active by fibrin- plate/ELISA	Recombinant BHK
QuickGO / UniProt (K9IJK6, P00750)	Computational / database	Refutes retention	Are the four terms experimentally supported for K9IJK6?	All four IEA-only; three TreeGrafter terms share one with-ref PANTHER:PTN000667065; finger/EGF exosites absent	Annotation provenance

GO Curation Implications

Lead requiring curator verification. The evidence separates cleanly into a supported core and an over-annotated periphery.

GO term	Aspect	Current evidence on K9IJK6	Recommended action (lead)
GO:0004252 serine-type endopeptidase activity	MF	Intact triad; DSPA-gamma amidolytically active (P 7592732)	Retain (core MF)
GO:0031639 plasminogen activation	BP	Fibrin-dependent activation demonstrated (P 7592732) (P 1309059)	Retain (core BP); IEA basis but strongly literature-consistent
GO:0005576 extracellular region	CC	Signal peptide; salivary secreted (P 23411029)	Retain (core CC)
GO:0048008 PDGF receptor signaling pathway	BP	IEA:TreeGrafter only; indirect/inefficient/context-specific in tPA; exosites absent	Remove or demote to non-core
GO:0014909 smooth muscle cell migration	BP	IEA:TreeGrafter only; downstream ECM-proteolysis effect in vessel wall	Remove or demote to non-core
GO:0033993 / GO:0071396 (cellular) response to lipid	BP	IEA:ARBA only; separate rule propagation; no direct evidence	Remove or demote to non-core
GO:1901701 cellular response to oxygen-containing compound	BP	IEA:ARBA only; very generic; no direct evidence	Remove or demote to non-core

The four disputed terms are **BP** annotations propagated electronically. They are neither anchored in a direct molecular activity of DSPA-gamma nor supported by any experiment on this accession or any DSPA isoform. Because they derive from a single grafted PANTHER node (three terms) plus generic ARBA rules (two terms), and because the structural exosites and biological context they depend on are absent, they fit the classic profile of **paralog/pathway**

over-annotation and should not be treated as core functions. No "protein binding" fallback is recommended; the informative, supported terms are the protease and plasminogen-activation terms above.

Mechanistic Scope

The **immediate molecular function** under test is enzymatic: a chymotrypsin-like serine protease (KP architecture) that, in a fibrin-dependent manner, cleaves plasminogen to plasmin. That direct activity is supported.

The four disputed terms describe **downstream phenotypes and pathway consequences**, not direct gene-product activities:

- **PDGFR signaling** is two steps removed — tPA cleaves latent PDGF-CC, and PDGF-CC (not tPA) engages the receptor; the cleavage is inefficient and cofactor-dependent.
- **Smooth muscle cell migration** is a tissue-level outcome of ECM proteolysis generating chemotactic gradients in the vessel wall.
- **Response to lipid** and **cellular response to oxygen-containing compound** are broad regulatory/environmental-response categories propagated by sequence rules, not molecular activities of the protein.

For a secreted bat salivary anticoagulant, none of these downstream/context-dependent processes is a plausible core function, and none has been demonstrated.

Conflicts and Alternatives

1. **Paralog/ortholog carry-over (primary alternative explanation).** The three TreeGrafter terms share the identical with-reference PANTHER:PTN000667065 (a eutherian tPA subtree), indicating en-bloc grafting rather than term-specific evidence. This is the most parsimonious explanation for their presence and directly competes with the "retention" framing.
2. **Full-length tPA vs. DSPA-gamma conflation.** The disputed functions are documented for full-length human/mouse tPA operating in the vascular wall and brain. DSPA-gamma lacks the finger and EGF domains and the mammalian tissue context. The seed hypothesis itself

warns against this conflation, and the evidence confirms the warning cuts *against* retention of the four terms.

3. **Organism/context mismatch.** tPA's PDGF-CC/BBB and SMC-migration roles are intravascular/CNS phenomena

(
P 18568034
P 16363896
P 23218119
).

A salivary secretion delivered into a host bite wound has no access to that machinery.

4. **In-vitro-only / inefficient reaction.** Even for tPA, PDGF-CC activation is "very inefficient" in vitro and cofactor-dependent

(
P 28725968
),

weakening any argument that the activity is a robust, transferable core function.

5. **Separate ARBA propagation.** The lipid/oxygen terms come from a different automated channel (UniProtKB-ARBA, ECO:0000256) with its own rule scope; they should be evaluated independently and are even more generic/weakly anchored.

No evidence was found that *supports* retention of the four terms as direct functions.

Limitations and Knowledge Gaps

- **No direct DSPA-gamma functional assays exist** for PDGF-CC cleavage, PDGFR signaling, SMC migration, or lipid/oxygen responses. We can demonstrate absence of positive evidence and absence of key structural determinants, but not experimentally proven loss of function. *Why it matters:* curators distinguish "unsupported" from "refuted"; this is the former, strengthened by structural and contextual arguments. *Resolution:* a direct comparative assay (below).
- **PDGF-CC cleavage by DSPA-gamma has never been tested.** tPA's protease domain performs the cleavage; DSPA-gamma has a homologous, active protease domain but different exosites and strict fibrin dependence. Whether the isolated catalytic domain could cleave PDGF-CC in vitro is unknown. *Resolution:* recombinant DSPA-gamma + latent PDGF-CC cleavage assay.

- **Sequence-level exosite mapping was based on UniProt/domain annotations, not a solved DSPA-gamma structure.** The finger/EGF absence is unambiguous (domains are simply not present), but fine-grained protease-domain surface-loop differences were not modeled.
- **ARBA rule identity confirmed but rule logic not fully audited.** The specific motifs triggering GO:0071396 and GO:1901701 were not dissected; these terms are generic and low-information regardless.

Discriminating Tests

1. **Direct comparative plasminogen-activation panel (DSPA-gamma vs DSPA-alpha1 vs tPA).** Recombinant expression (BHK secretion is established,

P 1634121

followed by kinetic assays $\pm$ fibrin to confirm the core function and benchmark fibrin dependence — anchors the retained terms.

2. **PDGF-CC cleavage assay.** Incubate purified recombinant DSPA-gamma with latent PDGF-CC $\pm$ Mac-1/LRP1 cofactors; SDS-PAGE/Western for the activated fragment. A negative result would strongly refute GO:0048008 propagation; a positive (even if inefficient) would partially rehabilitate an indirect link.
 3. **SMC migration / ECM-degradation assay.** Boyden chamber or ECM-gradient assay with DSPA-gamma vs tPA to test whether the bat enzyme can generate chemotactic ECM cleavage in the absence of finger/EGF-mediated cell-surface localization.
 4. **Annotation-provenance audit at scale.** Confirm across the DESRO gene set how many process terms trace to PANTHER:PTN000667065 grafts, to identify systematic tPA-paralog over-annotation.
 5. **Structural modeling / superposition** of a DSPA-gamma model against tPA (P00750) to quantify exosite/loop divergence around the protease active site.
-

Curation Leads (require curator verification)

Action changes (leads): - **Retain** GO:0004252 (serine-type endopeptidase activity), GO:0031639 (plasminogen activation), GO:0005576 (extracellular region) as core, literature-consistent annotations. - **Remove or mark as non-core / over-annotated** GO:0048008 (PDGF receptor signaling), GO:0014909 (smooth muscle cell migration), GO:0033993/GO:0071396 (response to lipid), GO:1901701 (cellular response to oxygen-containing compound). Rationale to record: IEA carry-over from PANTHER:PTN000667065 (three terms) and ARBA rules (two terms); downstream, cofactor-dependent, tissue-context-specific tPA biology; finger/EGF exosites absent from the target.

Candidate references with exact snippets to verify: -

P 7592732

— "All DSPAs are single-chain molecules, displaying substantial amidolytic activity." and "...fibrin selectivity... of DSPA alpha 1, alpha 2, beta, gamma, and t-PA was found to be 13,000, 6500, 250, 90, and 72, respectively." → supports core MF/BP. -

P 1937019

— "DSPA beta and -gamma lack the F and F-EGF domains, respectively." → supports KP architecture / exosite loss. -

P 18568034

— "PDGF-CC is a downstream substrate of tPA within the neurovascular unit." → supports demotion of GO:0048008 (indirect). -

P 28725968

— "in vitro, activation of PDGF-CC by tPA is very inefficient..." → supports demotion of GO:0048008. -

P 16363896

— "...tPA can generate cleaved extracellular matrix (ECM) chemotactic gradients to guide the migration of vascular smooth muscle cells (SMCs)." → supports demotion of GO:0014909 (downstream). -

P 23411029

— Vampirome salivary transcriptome/proteome → supports GO:0005576, salivary identity of JAA47048.1/K9IJK6.

Suggested curator questions: - Should IEA process terms grafted en bloc from a single tPA PANTHER node be retained on a domain-reduced salivary paralog lacking the mediating exosites? - Do we distinguish "unsupported/over-annotated" (this case) from "refuted" in the review action, given no direct negative assay exists?

Suggested experiments: the DSPA-gamma vs alpha1/tPA comparative panel, PDGF-CC cleavage assay, and SMC-migration assay described under Discriminating Tests.

K9IJK6 (DSPA-gamma) GO annotation curation assessment — all process terms IEA (electronic)				
GO_ID	Term	Aspect	Evidence	Assessment
GO:0004252	serine-type endopeptidase activity	MF	IEA:UniProtKB-EC	SUPPORTED (core MF) - retain
GO:0031639	plasminogen activation	BP	IEA:TreeGrafter	SUPPORTED (core BP) - retain
GO:0005576	extracellular region	CC	IEA:UniProtKB-SubCell	SUPPORTED - retain
GO:0048008	PDGF receptor signaling pathway	BP	IEA:TreeGrafter	OVER-ANNOTATED - remove/non-core
GO:0014909	smooth muscle cell migration	BP	IEA:TreeGrafter	OVER-ANNOTATED - remove/non-core
GO:0033993	response to lipid	BP	IEA:UniProtKB-ARBA	WEAK - remove/non-core
GO:1901701	cellular response to oxygen-containing co	BP	IEA:UniProtKB-ARBA	WEAK - remove/non-core

Figure 2. GO decision table, tPA-vs-DSPA-gamma domain comparison, catalytic-triad confirmation (His189/Asp238/Ser345), and annotation provenance for K9IJK6.

Proposed Follow-up Actions

- 1. Update the review** to retain the protease/plasminogen-activation/extracellular core and flag the four disputed process terms as IEA over-annotation candidates for removal or non-core demotion, citing the provenance (single PANTHER:PTN000667065 graft + ARBA rules) and the finger/EGF exosite absence.
- 2. Record the exosite comparison** (finger 39–81 / EGF 82–120 present in tPA, absent in K9IJK6) as structural justification in the review notes.
- 3. Escalate the systematic issue:** audit other DESRO/bat salivary PA entries for the same PANTHER:PTN000667065 en-bloc graft to catch analogous over-annotations.
- 4. If experimental resolution is desired,** prioritize the PDGF-CC cleavage assay with recombinant DSPA-gamma as the single most discriminating test.

Bottom Line

The seed hypothesis is **partially supported**: its insistence on treating K9IJK6 as the compact DSPA-gamma (not tPA/desmoteplase) and on not misreading the PRU00121 kringle caution as catalytic death is correct, and the protease/plasminogen-activation core is directly supported and should be retained. But the specific claim that K9IJK6 **retains** PDGF receptor signaling, smooth muscle cell migration, response to lipid, and cellular response to oxygen-containing compound as gene-product functions is **not supported** — these are IEA-only, en-bloc-grafted (PANTHER:PTN000667065) or generic-ARBA carry-overs of full-length mammalian tPA's downstream, cofactor-dependent, tissue-specific roles, and the mediating finger/EGF exosites are physically absent from the target. **Verdict: partially supported / over-annotated.** Recommended curation lead: retain the protease, plasminogen-activation, and extracellular-region terms; remove or demote the four disputed process terms to non-core pending direct DSPA-gamma assays.

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery